

**SIMATS ENGINEERING
ASSESSMENT TOOL 1**

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1709

CO1-AT1-Analytical Problem Solving

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COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 40%

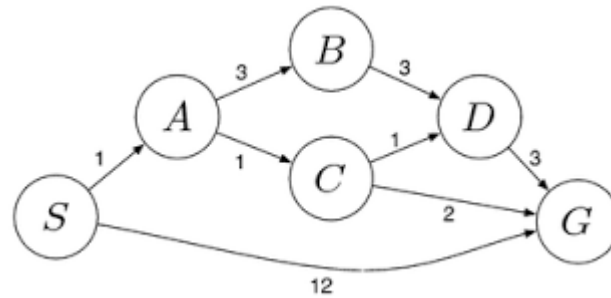
QUESTIONS: 5

Total Marks: 50

1. Solve the Water Jug problem: you are given 2 jugs, a 4-gallon one and 3-gallon one. Neither has any measuring maker on it. There is a pump that can be used to fill the jugs with water. How can you get exactly 2 gallons of water into 4-gallon jug? Explicit assumptions: A jug can be filled from the pump, water can be poured out of a jug onto the ground, water can be poured from one jug to another and that there are no other measuring devices available.
2. Find out how Mars Rover, analyse and explore the samples on Mars by answering the following questions.
 - i. What are the percept's for this agent?
 - ii. Characterize the operating environment.
 - iii. What are the actions the agent can take?
 - iv. How can one evaluate the performance of the agent?
 - v. What sort of agent architecture do you think is most suitable for this agent? Why?
3. Explore the search space using search strategies and formulate the problem components for the 8-Queens problem using the following information: Place 8 queens on a chessboard such that none of the queen's attack any of the others. A configuration of 8 queens on the board as shown in the figure below, but this does not represent a solution as the queen in the first column is on the same diagonal as the queen in the last column.
4. A customer wants to travel from the one location to another location using OLA Cab booking mobile application. The customer can select the any of the cab type as mini, micro, sedan, shared, prime and so on based on his comfort to reach the destination. Identify what type of problem-solving agent can be used and write the pseudocode for the above problem.
5. A logistics company operates a delivery network where packages must be transported between warehouses at minimum cost. Each route between warehouses has an associated travel cost (fuel + time). The company wants to determine the least-cost

path from the starting warehouse S to the destination warehouse G using the Uniform Cost Search (UCS) algorithm.

The delivery network is represented as a weighted graph:



Code of Conduct:

I V.Sireesha(192472208) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

V.Sireesha

RUBRICS FOR CALCULATION:

Criterion	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Max Marks
Problem Understanding	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem	10
Method Selection	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method	10
Solution Process	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process	12.5
Accuracy of Calculation	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations	10
Interpretation of Results	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation	7.5

1. Water Jug Problem

Problem Understanding

The problem requires obtaining exactly 2 gallons of water in the 4-gallon jug using only a 4-gallon jug, a 3-gallon jug, and a water pump. The allowed operations are filling a jug, emptying a jug, and pouring water from one jug to another. This is a state-space search problem, where each state represents the amount of water in both jugs.

Method Selection

The appropriate method is State Space Search (Uninformed Search) because the problem can be represented as a set of states and transitions. Each action changes the current state until the goal state is reached. This method guarantees finding a valid sequence of operations.

Solution Process

State $\rightarrow (x,y)$

$x \rightarrow$ 4 gallon

$y \rightarrow$ 3 gallon

$x=0,1,2,3,4$ and $y=0,1,2,3$

start state : (0,0)

Goal state: (2,n) for any n.

- (0,0) $\rightarrow$ Start state
- (0,3) $\rightarrow$ Rule 2 , fill the 3 gallon jug
- (3,0) $\rightarrow$ Rule 9 , pour all the water from 3 gallon jug to 4 gallon jug
- (3,3) $\rightarrow$ Rule 2 , fill the 3 gallon jug
- (4,2) $\rightarrow$ Rule 7, pour water from 3 gallon jug to 4 gallon jug until the 4 gallon jug is full
- (0,2) $\rightarrow$ Rule 5, empty the 4 gallon jug on the ground
- (2,0) $\rightarrow$ Rule 9, pour all water fro 3 gallon jug to the 4 gallon jug
- Goal state reached.

Accuracy of Calculation

After performing all operations correctly, the 4-gallon jug contains exactly 2 gallons, satisfying the problem requirement.

Final State

- 4-Gallon Jug = 2 gallons
- 3-Gallon Jug = 0 gallons

Interpretation of Results

The sequence of operations successfully reaches the goal state without using any measuring device other than the two jugs. The solution demonstrates how state-space search can solve AI problem-solving tasks systematically and efficiently.

2. Mars Rover

Problem Understanding

The Mars Rover is an intelligent autonomous agent used to explore the surface of Mars. It collects information about rocks, soil, atmosphere, and climate by using different sensors and cameras. Since human control is delayed due to the long communication distance between Earth and Mars, the rover must make decisions independently. The objective is to safely navigate the Martian surface, collect scientific samples, analyze them, and send the results back to Earth.

Method Selection

The most suitable AI method is a Goal-Based and Utility-Based Intelligent Agent.

- A Goal-Based Agent plans actions to achieve mission objectives such as reaching a target location or collecting a sample.
- A Utility-Based Agent chooses the best action by considering factors such as battery power, travel time, terrain difficulty, and scientific importance.
- This combination enables the rover to work efficiently in an uncertain and dynamic environment.

Solution Process

i. What are the percepts for this agent?

The Mars Rover collects information from its surroundings using sensors and cameras. It receives images of the surface, detects rocks and soil, measures temperature and atmospheric conditions, checks its battery level, and identifies obstacles while moving.

ii. Characterize the operating environment.

The Mars Rover works in a partially observable environment because it cannot see the entire area at once. The environment is dynamic due to changing weather and terrain. It is sequential because each action affects the next step, and uncertain because unexpected obstacles may appear.

iii. What actions can the agent take?

The Mars Rover can move forward, backward, left, or right, capture images, collect rock and soil samples, analyze the samples, avoid obstacles, and transmit the collected information back to Earth.

iv. How can one evaluate the performance of the agent?

The rover's performance is evaluated by checking whether it successfully reaches the target location, collects and analyzes accurate samples, avoids obstacles, uses battery power efficiently, and sends reliable scientific data to Earth.

v. What sort of agent architecture is most suitable? Why?

A Goal-Based and Utility-Based Agent is the most suitable architecture. It helps the rover achieve mission goals by selecting the best possible action based on the environment, available resources, and mission priorities. This allows the rover to work efficiently and safely without continuous human control.

Accuracy of Calculation

This question does not involve numerical calculations. The solution is accurate because it correctly identifies the Mars Rover's percepts, operating environment, actions, performance measures, and suitable agent architecture based on the principles of Artificial Intelligence.

Interpretation of Results

The Mars Rover is an excellent example of an intelligent autonomous agent. It continuously perceives its environment through sensors, processes the collected information, and performs suitable actions to achieve its exploration goals. Using a Goal-Based and Utility-Based architecture enables the rover to make efficient decisions, safely navigate the Martian surface, conserve energy, and maximize scientific discoveries. This demonstrates the practical application of AI agents in real-world space exploration.

3.8-Queens Problem

Problem Understanding

The 8-Queens problem is a classic Artificial Intelligence search problem. The objective is to place 8 queens on an 8×8 chessboard so that no two queens attack each other. A queen can attack another queen if they are in the same row, same column, or same diagonal. The task is to find a valid arrangement where all eight queens are placed safely.

Method Selection

The most suitable method is State Space Search using Backtracking (Depth First Search).

Reason:

- The problem has many possible arrangements.
- Backtracking places one queen at a time.
- If a queen attacks another queen, the algorithm returns to the previous step and tries another position.
- This reduces unnecessary searching and efficiently finds a valid solution.

Solution Process

Problem Formulation

Initial State

- An empty 8×8 chessboard with no queens placed.

State Space

- Every possible arrangement of queens on the chessboard.
- Each state represents a partial or complete placement of queens.

Actions

- Place one queen in a safe position in the next column.
- Continue placing queens until all 8 queens are placed.

Goal Test

- All 8 queens are placed.
- No two queens share the same row, column, or diagonal.

Path Cost

- Every move has the same cost.
- The objective is to reach a valid solution rather than minimize cost.

Search Strategy

1. Start with an empty chessboard.
2. Place the first queen in the first column.
3. Move to the next column and place another queen in a safe row.
4. Continue placing queens one by one.
5. If no safe position is available, backtrack to the previous column.
6. Change the previous queen's position and continue searching.
7. Repeat until all eight queens are placed safely.

One Valid Solution

Column	1	2	3	4	5	6	7	8
Row	1	5	8	6	3	7	2	4

This means:

- Queen 1 → Row 1, Column 1
- Queen 2 → Row 5, Column 2
- Queen 3 → Row 8, Column 3
- Queen 4 → Row 6, Column 4
- Queen 5 → Row 3, Column 5
- Queen 6 → Row 7, Column 6
- Queen 7 → Row 2, Column 7
- Queen 8 → Row 4, Column 8

Accuracy of Calculation

The above arrangement satisfies all the constraints:

- No two queens are in the same row.
- No two queens are in the same column.
- No two queens are on the same diagonal.

Interpretation of Results

The 8-Queens problem demonstrates how AI search algorithms solve constraint satisfaction problems. Backtracking explores only safe positions and eliminates invalid arrangements, making the search efficient. This technique is widely used in scheduling, resource allocation, puzzle solving, and optimization problems.

4. OLA Cab Booking Mobile Application

Problem Understanding

The problem involves a customer who wants to travel from one location to another using the OLA Cab Booking mobile application. The customer can choose different cab types such as Mini, Micro, Sedan, Shared, or Prime based on comfort, budget, and travel requirements. The AI system should identify the most suitable cab, estimate the fare, assign a nearby driver, and complete the booking efficiently.

Method Selection

The most suitable problem-solving agent is a Goal-Based Agent.

Reason:

- The customer's goal is to reach the destination safely and comfortably.
- The agent selects the best cab based on the user's preferences.
- It makes decisions to achieve the desired goal, such as finding the nearest available driver and booking the selected cab.

Solution Process

Step 1: The customer opens the OLA application and enters the pickup location and destination.

Step 2: The system searches for available cab types such as Mini, Micro, Sedan, Shared, and Prime.

Step 3: The application displays the estimated fare, arrival time, and available drivers.

Step 4: The customer selects the preferred cab based on comfort and budget.

Step 5: The system confirms the booking and assigns the nearest available driver.

Step 6:The customer tracks the driver's location until pickup.

Step 7:After reaching the destination, the payment is completed and the trip ends.

Pseudocode

```
Start
Input Pickup_Location
Input Destination
Display Available Cab Types
User selects Cab Type
Calculate Fare
Find Nearest Driver
If Driver Available Then
    Confirm Booking
    Assign Driver
    Track Trip
    Complete Payment
Else
    Display "No Cab Available"
End If
Stop
```

Accuracy of Calculation

The algorithm correctly performs the following operations:

- Accepts pickup and destination locations.
- Displays available cab options.
- Calculates the estimated fare.
- Assigns the nearest available driver.
- Completes the trip successfully.
- Displays a message if no cab is available.

Thus, the goal of booking a suitable cab is achieved accurately.

Interpretation of Results

The Goal-Based Agent successfully helps the customer reach the destination by selecting the appropriate cab based on the user's preferences. It automates the booking process, reduces waiting time, provides fare estimation, and improves the overall user experience. This demonstrates the practical application of AI agents in real-world transportation systems.

5.Uniform Cost Search (UCS) – Logistics Company

Problem Understanding

A logistics company wants to transport packages from the starting warehouse (S) to the destination warehouse (G) at the minimum possible cost. Each route between warehouses has a travel cost (fuel + time). The objective is to find the least-cost path using the Uniform Cost Search (UCS) algorithm.

Method Selection

The most suitable method is Uniform Cost Search (UCS).

Reason:

- UCS expands the node with the lowest cumulative cost first.
- It always finds the optimal (least-cost) path when all edge costs are positive.
- It is widely used in route planning, logistics, and navigation systems.

Solution Process

Step 1: Start from the source warehouse S.

Step 2: Insert S into the priority queue with cost 0.

Step 3: Remove the node with the lowest cumulative cost from the priority queue.

Step 4: Expand all neighboring warehouses and calculate their cumulative travel costs.

Step 5: Add the neighboring nodes to the priority queue and sort them according to the lowest cost.

Step 6: Repeat the process by expanding the node with the smallest cumulative cost.

Step 7: Continue until the destination warehouse G is reached.

Step 8: Display the least-cost path and its total travel cost.

Pseudocode

Start

Insert Source (S) into Priority Queue with Cost = 0

While Queue is not Empty

 Remove Node with Lowest Cost

 If Node = Goal (G)

 Display Least Cost Path

 Stop

 Expand Neighboring Nodes

 Calculate New Cost

 Insert Neighbor into Priority Queue

End While

Stop

Accuracy of Calculation

- UCS always expands the node with the minimum cumulative cost.
- It guarantees the optimal path when all edge costs are positive.
- The final answer consists of:
 - The least-cost path from S to G.
 - The minimum total travel cost.

Interpretation of Results

Uniform Cost Search efficiently finds the minimum-cost route between warehouses by always choosing the path with the lowest accumulated cost. This reduces transportation expenses, improves delivery efficiency, saves fuel and time, and helps logistics companies make better routing decisions. UCS is widely used in GPS navigation, transportation, robotics, and supply chain management because it guarantees the optimal solution.